

Final 23-02-2022

- ① 3 trozos de long $L = 1\text{m}$ y $\lambda_1 = \frac{80\text{W}}{\text{mk}}$; $\lambda_2 = \frac{50\text{W}}{\text{mk}}$; $\lambda_3 = \frac{40\text{W}}{\text{mk}}$

Por extremos están a T_1 y T_2 .

Reg. Estacionario $\frac{H}{S} = 100 \frac{\text{W}}{\text{m}^2}$; $T_2 = 300\text{K}$

Si $T_1 > T_2$, Calcular T_1 .

$$H = \lambda S \frac{\Delta T}{L}$$



$$\frac{H_1}{S} = \frac{H_2}{S} = \frac{H_3}{S}$$

$$\frac{H}{S} = \lambda \frac{\Delta T}{L}$$

$$\lambda_1 \frac{(T_1 - T_{u1})}{L} = \lambda_2 \frac{(T_{u1} - T_{u2})}{L} = \lambda_3 \frac{(T_{u2} - T_2)}{L}$$

$$\frac{H_3}{S} = \lambda_3 \frac{(T_{u2} - T_2)}{L}$$

$$\frac{100 \frac{\text{W}}{\text{m}^2}}{\frac{40 \text{W}}{\text{mk}}} = T_{u2} - T_2 \Rightarrow 2,5\text{K} + 300\text{K} = T_{u2}$$

$$T_{u2} = 302,5\text{K}$$

$$\frac{H_2}{S} = \lambda_2 \frac{(T_{u1} - T_{u2})}{L} = \frac{100 \text{W}}{\text{m}^2} \Rightarrow T_{u1} - T_{u2} = 2\text{K}$$

$$T_{u1} = 304,5\text{K}$$

$$\Rightarrow \frac{100 \text{W}}{\text{m}^2} = \lambda_1 \frac{(T_1 - T_{u1})}{L} \Rightarrow 1,25 = T_1 - T_{u1}$$

$$T_1 = 305,75\text{K}$$

- ② Um mol de gas ideal a $T_A = 300\text{K}$ y $P_A = 100\text{kPa}$ se calienta a Vete hasta $P_B = 200\text{kPa}$ y luego se reduce el volumen hasta $T_C = 300\text{K}$. Calcular ΔU_{CA}

$$\Delta U_{CA} = C_V n (T_A - T_C)$$

$$\Delta U_{CA} = 0$$

- ③ 2 Esf. conductores de $R_1 = 9 \text{ cm}$ y $R_2 = 3 \text{ cm}$ separados por di mayor a sus radios con $V_1 = 3000 \text{ V}$ y $V_2 = 5000 \text{ V}$ resp de ∞ . Se los conecta con un alambre conductor hasta alcanzar estado electrostático. Calcular el campo de c/ esfera luego de la conexión.

$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{R_1} = 3000 \text{ V}$$

$$V_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{R_2} = 5000$$

$$q_1 = 3000 \text{ V} \cdot R_1 \cdot 4\pi\epsilon_0$$

$$q_2 = 5000 \cdot R_2 \cdot 4\pi\epsilon_0$$

$$q_1 = 30 \text{ nC}$$

$$q_2 = 16,7 \text{ nC}$$

Después el V se iguala.

$$V_1 = V_2 \Rightarrow \frac{1}{4\pi\epsilon_0} \frac{q_1}{R_1} = \frac{1}{4\pi\epsilon_0} \frac{q_2}{R_2} \Rightarrow \frac{q_1}{9} = \frac{q_2}{3} \Rightarrow \frac{q_1}{3} - \frac{q_2}{3} = 0$$

$$q_1 + q_2 = q_1 + q_2$$

$$46,7 \text{ nC} = q_1 + q_2 \quad (2)$$

$$q_1 = 35 \text{ nC}$$

$$q_2 = 11,6 \text{ nC}$$

④

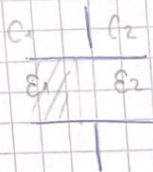
$$A = 500 \text{ cm}^2$$

$$d = 4 \text{ mm}$$

El 40% del Volumen es

$$\epsilon_r = 4$$

Se conecta a 50V. Calcular ΔV si se retira el dieléctrico y con la fuente desconectada



Paralelo

$$A = 0,05 \text{ m}^2$$

$$\Rightarrow C_{eq} = C_1 + C_2$$

$$C_1 = \frac{\epsilon_0 \cdot \epsilon_r \cdot A}{d} = 1177,08 \text{ pF}$$

$$C_2 = \frac{\epsilon_0 \cdot A}{d} = 66,3 \text{ pF}$$

$$C_{eq} = 0,243 \text{ nF}$$

$$Q_T = C_{eq} \cdot 50 \text{ V} = 12,17 \text{ nC}$$

Al sacar la batería la carga se mantiene. Calcular nuevo C_{eq}

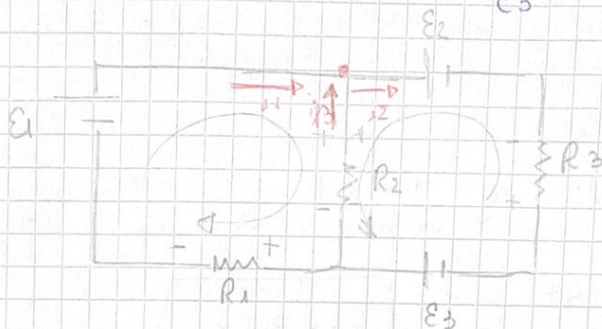
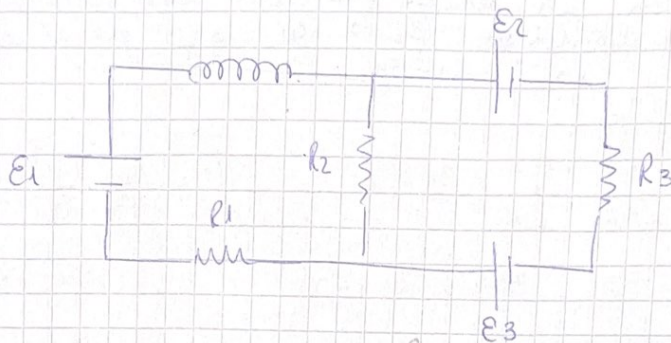
$$C_{eq2} = \frac{\epsilon_0 \cdot A}{d} = 0,1106 \text{ nF}$$

$$\Rightarrow \Delta V = \frac{Q_T}{C_{eq2}} \Rightarrow \Delta V = \frac{12,17 \text{ nC}}{0,1106 \text{ nF}} = 110 \text{ V}$$

5) Reg. Exercicio calcular Potencia dissipada em R_2

$$\mathcal{E}_1 = 25V \quad \mathcal{E}_2 = 12V \quad \mathcal{E}_3 = 12V$$

$$R_1 = 20\Omega, \quad R_2 = 10\Omega \quad R_3 = 10\Omega \quad L = 446mH$$



$$\textcircled{1} \quad \mathcal{E}_1 - i_1 R_2 - i_1 R_1 - i_2 R_2 = 0$$

$$\textcircled{2} \quad -\mathcal{E}_3 - i_2 R_3 - \mathcal{E}_2 - i_2 R_2 - i_1 R_2 = 0$$

$$\Rightarrow 25V - i_1 10 - i_1 20 - i_2 10 = 0$$

$$25V - 30i_1 - 10i_2 = 0$$

$$-30i_1 - 10i_2 = -25V$$

$$\textcircled{2} \quad -12 = 20i_2 - 10i_1 - 12 = 0$$

$$-10i_1 - 20i_2 = 24$$

$$i_1 = \frac{37}{25}$$

$$i_2 = -\frac{91}{50}$$

$$i_1 + i_3 = 12$$

$$i_1 = 1,48A$$

$$i_2 = 1,94A$$

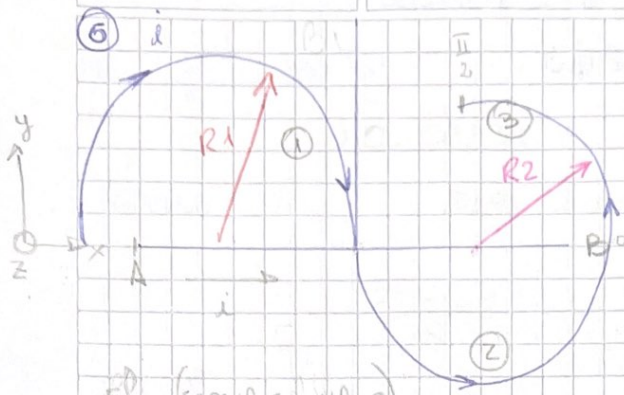
$$i_3 = 0,46A$$

$$P_{R_2} = i_3^2 \cdot R_2 = 2,116W$$

$$B = 5 \text{ (k) T}$$

$$R_1 = 2 \text{ m} \quad R_2 = 1,5 \text{ m}$$

FECHA



$$i = 3 \text{ A}$$

Calcular la fuerza
sobre el circuito

$$F_{\text{mag}} = i \int dl \times B$$

$$F_{\text{mag T}} = F_{\text{mag 1}} + F_{\text{mag 2}} + F_{\text{mag 3}}$$

$$F = i \cdot 5r \int \cos \theta; \sin \theta; 0 \, d\theta$$

$$F_1 = 3 \cdot 5 R_1 (\sin \theta; -\cos \theta; 0) \Big|_0^\pi$$

$$F_1 = 30 \cdot (0; -2; 0) = -60 \hat{j}$$

$$F_2 = 3 \cdot 5 R_2 (\sin \theta; -\cos \theta; 0) \Big|_0^\pi$$

$$F_2 = 22,5 (0; -2; 0) = -45 \hat{j}$$

$$F_3 = 3 \cdot 5 R_2 (\sin \theta; -\cos \theta; 0) \Big|_0^{\pi/2}$$

$$F_3 = 22,5 (1; 1; 0) = 22,5 \hat{i} + 22,5 \hat{j}$$

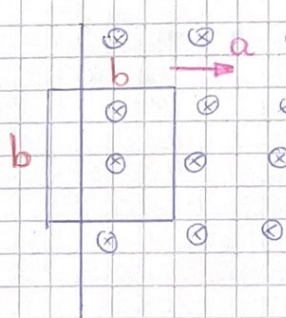
$$F_T = 22,5 \hat{i} - 81,5 \hat{j}$$

7 Espira cuadrada de lado $b = 10 \text{ cm}$ en el xy parte del espacio
con $a = 3 \text{ m}$. Ingreso en un B uniforme y cte. $B = -10 \text{ k T}$

a) Hallar $\mathcal{E}$ electromotriz inducida (fem) en la espira en
función del tiempo para $0 < x < b$

$$\mathcal{E} = -L \frac{di}{dt}$$

b) Escalar expresión que permite calcular i si $L = 2 \text{ mH}$ y $R = 100 \Omega$



$$\mathcal{E} = -N \frac{d\Phi}{dt}$$

$$b dx$$

$$\Phi = \int B dA = B n \cos 180 dA$$

$$\Phi = -B \cdot b dx$$

$$\mathcal{E} = B b \frac{dx}{dt}$$

$$\mathcal{E} = B \cdot b \cdot v = B \cdot b \cdot at \quad |\mathcal{E}| = 3t$$

$$\text{b) } -\frac{dQ}{dt} = Ri(t) + L \frac{di(t)}{dt} \Rightarrow$$

$$3t = 100 \cdot i(t) + 2 \cdot 10^{-3} \frac{di(t)}{dt}$$

NOTA

8) $I_{ef} = 2A$. corrente adelanto $\frac{\pi}{3}$ con respecto a la tension . $f = 150 \text{ Hz}$

$$X_C = 800 \Omega \quad L = 0,6 \text{ H}$$

a) R

$$X_L = 2\pi 150 \cdot 0,6$$

b) V_{ef}

$$X_L = 180\pi$$

CIVIL

$$\text{tg}\left(-\frac{\pi}{3}\right) = \frac{X_L - X_C}{R}$$

CAPACITIVO

$$R = \frac{180\pi - 800}{\text{tg}\left(-\frac{\pi}{3}\right)}$$

$$R = 135,4 \Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \Rightarrow Z = 270,8$$

$$Z = \frac{V_{ef}}{I_{ef}} \Rightarrow V_{ef} = Z \cdot I_{ef}$$
$$V_{ef} = 541,58 \text{ V}$$